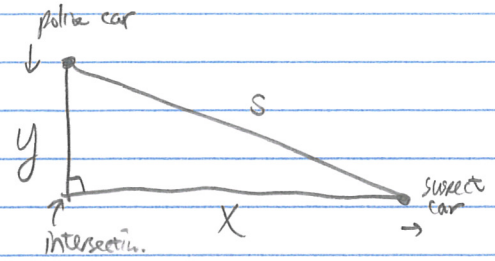
(3)



y = dist btwn police and intersection (y is decreasing)

x = dist btwn suspect and intersection (x is increasing)

why?

s = dist btwn the cars.

$$x^2 + y^2 = s^2$$

$$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 2s \cdot \frac{ds}{dt}$$

$x = 0.8$ $\quad y = 0.6$ $\quad \frac{dy}{dt} = -60$ $\quad \frac{ds}{dt} = 20$

$$(0.8)^2 + (0.6)^2 = s^2$$

$$s = \sqrt{(0.8)^2 + (0.6)^2}$$

$$2(0.8) \cdot \frac{dx}{dt} + 2(0.6) \cdot (-60) = 2\sqrt{(0.8)^2 + (0.6)^2} \cdot 20$$

$$2(0.8)\frac{dx}{dt} = 2(0.6)(60) + 40\sqrt{(0.8)^2 + (0.6)^2}$$

$$\frac{dx}{dt} = \frac{2(0.6)(60) + 40\sqrt{(0.8)^2 + (0.6)^2}}{12(0.8)} = 70$$

The suspect is moving east at a rate of 70 mph.